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Research article

Utilising law in the transition of the Kingdom of Saudi Arabia to a low-carbon economy

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ABSTRACT

The Kingdom of Saudi Arabia (KSA) has adopted a new law and policy framework, Vision 2030, designed to deliver a transition to a diversified and low-carbon economy. This paper analyses from an environmental law and policy perspective how law is providing opportunities and challenges in achieving the outcomes of this national masterplan. The paper employs traditional legal scholarship to assess the law itself and also explores its impact in practice, via an analysis of findings from expert stakeholder interviews. Key issues around environmental governance are discussed in depth in the light of Vision 2030. One of the key issues of focus is that of integration, in terms of the design and practice of law, institutions and outcomes. To date, it is clear that the further policies needed to ensure that environmental integration and a low-carbon transition take place are currently being created in the effort to deliver the aims of the Vision. This study finds that the transition to a low-carbon economy in the KSA is under-researched and under-explored in practice, from an environmental perspective. The findings attest that although some significant actions have been initiated, further action needs to be taken, given the importance of this transition, in the KSA in the light of the promising nature of Vision 2030 in this regard.

1. Introduction

The KSA is entering a new phase in its history, driven by a new national framework, known as Vision 2030. This national masterplan is premised on overarching and cross-cutting major themes that have direct implications on almost every aspect of the country's life, i.e. in terms of the economy, society, the environment and the law. According to its mission statement, "*Vision 2030 draws on the nation's intrinsic strengths. These strengths underpin the three themes that embody the Vision's ground-breaking agenda; A Vibrant Society - A Thriving Economy - An Ambitious Nation*". (Vision 2030: The Vision Themes, 2016).

One prime area of improvement that makes the study of the KSA important for societal transition scholars is the need to diversify the economy away from its heavy dependence on oil. Through Vision 2030, the KSA seeks to achieve transition to a lower-carbon economy, which entails diversifying the investment sources. For example, under the "Thriving Economy" theme, the Vision

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expressly states that "… *the Kingdom is using its investment power to create a more diverse and sustainable economy*" (Vision 2030: The Vision Themes). This includes developing the local renewable energy market through initiatives such as the King Salman Renewable Energy Initiative that is set to be achieved in Vision 2030. (The Vision Themes: Thriving Economy, investing for the future, 2016).

Scholars have drawn attention to the vital roles of law and policy in a societal transition, particularly in achieving a low-carbon transition (Heffron et al., 2018) and in the context of the evolution of the law in the energy transition (Heffron and Talus., 2016). Hence, there is a need to address environmental issues inherent in environmental law and policy. However, legislative development takes time, as a consensus has to be achieved across the spectrum of societal actors and new laws need to be in place so that investors will have the confidence to make the necessary investment. This is recognised by Vision 2030 in the form of a review of "legal and regulatory frameworks that allows the private sector to buy and invest in the renewable energy sector," which has already commenced. (The Vision Themes: Thriving Economy, investing for the future, 2016).

Leading climate scientists have warned that the transition to a low-carbon economy (or a sustainable economy as it is sometimes referred to) is generally happening at too slow a pace, worldwide (Figueres et al., 2017). Scholars note the key role that law (regulations and policy) can play in disrupting the system to ensure that the transition happens (Kivimaa et al., 2021). It has been long recognised that law has a role to play in transitions literature and yet there has been a lack of detailed exploration of this role and this is an area that this research aims to address. Although some recent studies such as that by Johnstone et al. (2020) do briefly highlight the role of law (regulations and policy) in low-carbon transitions specifically, it is not covered in the depth it deserves. This emphasis on the law perspective, utilising the terms 'governance' and 'justice' has been identified as a topic for further research in a key article by Köhler at al. (2019) exploring the agenda for sustainability transition research. In particular, they identify the need for more practice-based research, which the present article delivers in exploring the impact of law in practice and not just in its written, intended form. Moreover, legal research in the context of low-carbon or sustainable transitions also has a practitioner element inherent in it, as law provides the framework and legal certainty required for investors to make the investments necessary to realise the transition (Heffron et al., 2021).

In this context, low-carbon transitions are processes that involve risks as well as challenges (Vliet et al., 2020). This paper explores some of these elements in the context of environmental governance in the KSA, and the subsequent implications for the transition to a low-carbon economy, led by the recently launched national policy document Vision 2030.

The reforms laid out by Vision 2030 encapsulate the goal to "reduce all types of pollution" (Vision 2030: National Transformation Program, 2016). The intricacy of environmental problems requires environmental regulations and considerations to be integrated holistically in all areas of policy, whatever their main focus (Gabler, 2010), in order to "*avoid otherwise contradictory policy objectives*" (Bell and McGillivray, 2008, p.56). This holistic approach is referred to in law as the 'integration principle'. However, it is important to note that the principle can be interpreted in very different ways, including economic interpretations that focus on cost and price signals which are irrelevant to the discussion in this paper (Convery, 2010). Other scholars, for example, Perrson, 2007 have more radically questioned the qualification of "integration" and its content, arguing that its instinctive appeal masks great ambiguity. This paper examines the integration of policy as the KSA aims for its low-carbon transition. In essence, it asks to what degree environmental norms have been integrated into policy areas across different levels of governance" (Gabler, 2010). In this broad sense, the integration principle is a prerequisite for the real appreciation of the "wicked" and intricate character of environmental challenges, since protecting the environment in certain areas more than others or in specific environmental media or ecosystems will eventually result in environmental degradation.

This exploration of the uncharted territory of environmental and climate law and governance systems is a significant contribution of this paper. In this time of the COVID-19 pandemic, ensuring that these systems work, function and are integrated, is of vital importance, as is stressed in this paper. Further, utilising the KSA as an example is of significant value, since the economy of the state has long been dependent on fossil fuel energy production. Hence, these issues are even more important in the context of the current status of environmental governance in the KSA, and their implication for the transition to a low-carbon economy, led by Vision 2030.

The remainder of this paper is divided into three main sections. Following the account of the methodology in section two, section three reviews the principal environmental laws and policy documents in the KSA and explains the roles of the main actors and institutions involved. Section four focuses on the aims and content of Vision 2030 and their relationship with climate change issues. It also compares the understandings of these issues among local civil servants, academics and representatives from environmental societies and businesses, based on an analysis of semi-structured interviews carried out in this study. Section five explores Vision 2030 in terms of the practical steps required to achieve a low-carbon economy in KSA in the context of environmental law, while the conclusions and contribution of the paper are summarised in section six.

2. Research method

2.1. Introduction

A mixture of doctrinal and qualitative research methods was employed in each phase of the research, as each "legal" qualitative research method has its own strengths and shortcomings (Cane and Kritzer, 2010). Thus, a combination of valid methods will contribute to the validity and reliability and hence, in accordance with Patton's (1999) terminology, benefits from triangulation.

The empirical data were collected by undertaking fieldwork in the form of semi-structured interviews with the participants, which were conducted to obtain original insights into the subject matter and raise issues that have not been prominently covered or are not found within the currently available literature or in legal legislation and documentation from the KSA.

2.2. Doctrinal methodology

The black-letter-law approach was applied in this research and embedded in the qualitative discussion. The meaning of "doctrinal methodology" for the purpose of this research is simply to indicate a considerable reliance on legislations and legal instruments (Hutchinson and Duncan, 2012). Despite some criticism that it is "*rigid and inflexible*" compared to a more interdisciplinary approach (Vick, 2002), this doctrinal legal methodology is useful in extracting and identifying inherent principles, rationales and values from legal cases and legislation, since it is extensively predicated on raw legal materials, such as constitutional provisions, laws and judicial decisions (McConville & Chui, in McConville and Chui, 2007) As such, a range of legislation and legal instruments was considered in the research that underpins this paper, at various scales: internationally, regionally and nationally (the prime focus) as well as locally in certain major cities in the KSA. Resort to legal environmental cases and court judgments and decisions was, however, minimal, given that the publication of such legal environmental material is noticeably limited and not systematic in the KSA.

2.3. Qualitative research

This paper is mainly about a real-life phenomenon, with a focus on the KSA as a case study from the Middle East. Thus, among its main objectives is to provide an in-depth understanding of the current reality and practices of the main environmental governance institutions or actors in this context. The qualitative research methods used are explained below.

2.3.1. Empirical method: semi-structured interviews

Interviews have been proven useful in empirical law-focused research projects, since they allow an entry to the experts' insights and experience regarding the legal phenomenon being explored (Webley, 2010). As noted above, case study research should enable a deep understanding of certain phenomena or of the unit being investigated. One avenue to reach such this degree of understanding is not to rely solely upon a single type of qualitative data source, such as documents, although these may be a cornerstone, but actually to diversify the forms of data being utilised, in order to achieve depth of knowledge (Creswell, 2013). Here is where the importance of the interview as a data gathering strategy comes into this thesis. A key reason to utilise interviews in this research was that beside the recognised advantages of the interview method in case study projects (Yin, 2014), there is a lack of environmental governance data disclosed or documented by the relevant environmental entities.

Thus, to cope with this scarcity of environmental data and environmental governance studies on the KSA, primary data was generated via semi-structured in-depth interviews and analysed. The potential of the interview strategy in this regard is great, since it can "*reach areas of reality that would otherwise remain inaccessible.*" (Peräkylä and Ruusuvuori, 2011). Yin (2014, p. 106), for example, asserts the usefulness of interviews to gain deep understanding in case study research, if the researcher carries out his or her mission effectively and focuses on the "line of inquiry" and does not get deviated. To avoid deviation and facilitate the pursuit of the line of inquiry, an interview guide which was drawn up containing the set of question posed to the different interview respondents, as recommended by Patton (2002).

It was also borne in mind that when asking questions and pursuing points that it is important to pose the questions in an unbiased manner Yin (2014). The issue of the line of inquiry, as well as the issue of who to interview can be, and was, in this thesis, driven by the theoretical framework and literature review (regarding environmental governance), as well as by the research questions (Beitin, 2012). Further details on the interviewees and their selection are provided in Appendix A.

3. Environmental Law in the KSA in the context of a Low-Carbon Transition and the institutions involved in its implementation

3.1. Introduction

In order to achieve the low-carbon transition detailed in the Vision 2030 policy, the environmental law and institutions in KSA need to work in unison. This section briefly explains the background and development of the current environmental law and the institutions responsible for implementing it. It then explains the relationships among the institutions mandated to apply the law and the opportunities and challenges in these relationships which affect how it ultimately works in practice. The data from the interviews which were undertaken are also utilised to highlight how these opportunities and challenges relate to daily practice on the ground. Before analysing the environmental law of the context, it is worth highlighting that there are some institutional issues and developments which are further explored in more detail in Section 5.

3.2. The status of the general environmental law (GEL)

Although it is the principal environmental statute, the GEL (2001) and the equivalent legal instruments are yet to integrate and address climate change issues, which creates a significant gap for legal practice. Although climate change is dealt with by other "softer" instruments, such as The State of the Environment Report (2017) issued by GAMEP, the absence of any legally-binding targets in the GEL renders the contribution of the GEL in its current form less than promising in terms of achieving a transition to a low carbon economy.

These lacunae in the environmental legislature can be traced back to the general slow responsiveness by the environmental regulatory process. Procedures to respond to recent events and/or change over time are inhibited by cumbersome administrative

procedure and also the need for highly technical legislation; however, it should be noted this is common to many countries. A typical example is the GEL itself, which has not undergone any updating and review process since its release in 2001. With the KSA being a nation reliant on its energy resources, there has previously been little incentive to act quickly on environmental issues; however, that is now set to change with the advent of Vision 2030, and a transition to a low-carbon economy seems set to start. In this respect, there has been a considerable environmental legislative endeavour to address such gaps. This can be illustrated by the preamble of the new Environmental Law (2020) and the Cabinet Resolution No. 729 dated 16/11/1441 AH (i.e. July 2020), which clearly state that this new legislation and its various Rules for Implementations and By-laws (which are yet to be issued) will supersede the GEL (2001) and nine other wildlife conservation and environmental protection laws, regulations and resolutions.

These challenges have been addressed already by some European nations, notably by the UK. For example, the enactment of the UK Climate Change Act in 2008 transferred the challenge to the next step of practical challenges, including issues regarding judicial interaction in climate change issues and the potential of judicial review, as discussed in the UK context by Macrory (2014).

In addition, there are issues related to the complexity and polycentric nature of the problem of climate change per se. These are reflected in the formidable challenges facing lawyers and adjudicators in addressing climate-change-related legal disputes. These legal challenges and the "*legally disruptive nature of climate change*" (Fisher et al., 2017, p. 177) are not only relevant to the Middle East region or developing countries but are also encountered in high-income countries across the world.

The relative stagnancy of the legal arrangements around environmental and low carbon issues, has led to an established practice by several executive bodies, e.g. the Ministry of Energy (ME). This process involves having recourse to ministerial decisions and administrative orders in order to fill the legislative gap regarding many environmental issues, where the laws are either out-dated or simply do not exist, as demonstrated by the work undertaken in this regard by His Excellency the Minister of Energy (YouTube Channel, n.d.). The drawback of this sector-based approach is that it results in a non-integrated, sectoral and sporadic type of governance that is by nature neither comprehensive nor systematic, and thus incompatible with the goal of attaining a low carbon economy.

The KSA's main environmental regulator is the General Authority for Meteorology and Environmental Protection (GAMEP); however, this regulator has limits to its functions. These derive from a number of factors and the collective view of the interviewees was that these limitations relate to two main factors. The first of these factors is the parallel presence of decentralised environmental regulators, for example, the High Commission for the Development of Riyadh (HCDR) (now the Royal Commission for Riyadh City), and Royal Commission for Jubail and Yanbu (RCJY), with considerable power within their own zones. In addition, there is the presence of an additional agency, the Industrial Property Authority (MODON) which applies and implements the General Environmental Law (GEL) within its industrial cities. The second factor is GAMEP's limited capacity to intervene in regional issues. This was identified as a problem by participants from both the civil service and industrial sectors (A-1-3 and B-1-4). A participant from the civil service (A-1-8) suggested it would be considered contentious to intervene, and an interviewee from the civil service (I-A-10) also pointed out that while MODON is under GAMEP's general jurisdiction, it has its own environmental regulations, with regard to fiscal sanctions. This is interesting, because "legally" there is no reason to distinguish between the fiscal and non-fiscal provisions of the GEL by MODON, as the GEL is the principal and governing enactment. This situation demonstrates the insufficiency of the GEL to deliver proper environmental protection, which led MODON to come up with its own fiscal penalty guidance. The resulting division of authority can be very contentious, legally and validity-wise. Interestingly, as pointed out by (I-A), this issue has not been legally contested in the administrative courtroom or by any judicial means, which indicates the lack of judiciary involvement in environmental issues.

There is a balance to be achieved in terms of the institutions needed to bring about a low-carbon economy. For example, one of the challenges addressed below is that the main authority mandated to lead and work on the climate change file, both nationally and internationally, is the Ministry of Energy (ME), with less noticeable presence by fellow institutions such as GAMEP and the Ministry of Environment, Water and Agriculture (MEWA). This is in contrast to the challenges discussed in some western literature sources, where, for example, McEldowney (2016) notes that one of the challenges that complicates climate change affairs in the UK is their delegation to low regional and local levels. Hence, although a debate regarding how much institutional involvement is required, change and development is clearly needed and expected in the KSA towards the year 2030.

Therefore, updating the GEL, or even creating new legislation to encompass the issue of climate change and develop the capacity to deliver a low carbon economy would be advantageous in many respects, despite the short term economic costs incurred (Lazarus, 2008). It would enhance the harmonisation of the different priorities existing within the various ministries and government institutions and ensure institutional alignment of all the national entities with the common objective of securing a low-carbon economy. It would also open up new avenues for the courtrooms to become involved and to have a strong duty to review the effectiveness of new legislation; this is especially important for a legislation-based legal jurisdiction which is not a precedent-based or common law system.

3.3. The role of the 'Integration' principle in the GEL

Although the GEL is accompanied by an Appendix dictating the admissible amount of pollution to be emitted or released into the environment by industrial and development activities, there are a number of major omissions. Indeed, regulatory gaps still exist which are likely to undermine the entire environmental protection efforts exerted by stakeholders, primarily GAMEP. Such shortcomings appear to reflect either misunderstanding of the nature of the environmental and ecological systems and their ills, or at least a difficulty in coping with them. This is reflected in the focus (by the GEL and its Rules for Implementation) on air quality standards at the cost of integrating equally important issues such as the quality of water and soil, which are largely left unregulated. Similarly, the experts interviewed representing both civil servants and academics (1-A-1 and 1-C-2) pointed out that relatively few potential chemical emissions are addressed, with the rest untouched (I-A-4 and I-C-6).

Another related issue in the GEL is that it addresses specifically certain industrial activities, which include fertilizer plants, cement factories, and iron and steel plants, and expressly provides for their emissions and effluents. A participant from the bureaucracy (I-A-3) stated that, on a practical level, this specific articulation brings about cumbersome and unnecessary disputes by organisations involved in activities not mentioned above. Further, one of the participants from the bureaucracy (1-A-1) claimed that those involved in the activities not covered by the legislation actively explore the regulatory loopholes. Another interviewee from this group (1-A-3) put this down to the excessively legalistic approach and legislation-based style upon which GAMEP acts. Moreover, a third participant (1-A-10) pointed out that, even where GAMEP rejects a literal application of the GEL and employs its discretionary power, it is always prone to reversal where the regulated entity appeals or raises legal disputes. In addition, this legislation-based style of legal jurisdiction is also complicated by the lack of specialised environmentally-trained judiciary or tribunals (Table 1).

3.3.1. Opportunity to adapt the legal responsibility

The failure to integrate all the potential impacting activities and harmful emissions into decision making, and thus the law, makes it possible for the regulated polluting entities to "legally" manoeuvre their responsibility and to remain unaccountable. In addition, the focus of the environmental standards on gaseous emissions makes it possible, at least in theory, for industries to modify the process in order to release their pollutants in other ways, such as into the ground water and/or into the air. This is automatically aligned with the mainstream meaning of the integration principle, including the well-recognised core assumption by the Brundtland Report that "*the integration of environmental concerns into decision-making at the outset would enable policy decisions to contribute substantially to a transition toward SD* [sustainable development]" (Knudsen and Lafferty, 2016, p. 337).

3.3.2. Issues regarding 'integration' between economic sectoral activities

One of the major drawbacks of the GEL and thus the environmental governance in the context studied, is the vague or incomplete drafting of some of its provisions. This raises major practical problems, for example, the lacunae created by imposing an obligation without decisively pinpointing who is in charge of monitoring and ensuring its compliance. This creates a functional and operational vacuum in each public institution, as each potentially responsible institution might avoid the responsibility and cast it onto another. For example, Article 12 of the GEL places responsibility on contractors or owners for removal of waste or prevention of pollution and conforming to environmental regulation.

However, it is unclear who is in charge of ensuring compliance with this Article. Using the GEL terms, the licensing authority is "*any authority in charge of licensing projects of potentially adverse impact on the environment*" (GEL, Article 1, Item 4, 2001). This issue was

Table 1
Details of the Interviewees.

Category	Affiliation
First: Bureaucracy	
1 (I-A-1)	General Authority of Meteorology and Environmental Protection (GAMEP) - Ministry of Environment, Water and Agriculture.
2 (I-A-2)	(GAMEP) - Ministry of Environment, Water and Agriculture.
3 (I-A-3)	(GAMEP) - Ministry of Environment, Water and Agriculture.
4 (I-A-4)	(GAMEP) - Ministry of Environment, Water and Agriculture.
5 (I-A-5)	(GAMEP) - Ministry of Environment, Water and Agriculture.
6 (I-A-6)	(GAMEP) - Ministry of Environment, Water and Agriculture.
7 (I-A-7)	Ministry of Municipal and Rural Affairs.
8 (I-A-8)	Environmental Protection Department - Royal Commission for Jubail and Yanbu (RCJY)
9 (I-A-9)	Arriyadh Development Authority
10 (I-A-10)	The Saudi Industrial Property Authority (MODON) - Environmental Protection Department
Second: Industries/ Private Sector	
11 (I-B-1)	Oil and Petroleum Company
12 (I-B-2)	Power and Water Utility Company
13 (I-B-3)	Petrochemical Company
14 (I-B-4)	Petrochemical Company
15 (I-B-5)	Stone and Gravel Industry
16 (I-B-6)	Paint Factory
17 (I-B-7)	Aerospace and security company- Environmental Protection Department
18 (I-B-8)	Metal Coating Company
Third and Forth: Academia, Research Centres & Semi-NGOs	
19 (I-C-1)	King Abdullah Institute for Nanotechnology - Nanotechnology Society
20 (I-C-2)	King Saud University (KSU) -Society of Agricultural Sciences
21 (I-C-3)	King Abdulaziz University (KAU) - Environmental Science Society
22 (I-C-4)	King Abdulaziz University (KAU)
23 (I-C-5)	King Abdulaziz University (KAU) - Government Consultant
24 (I-C-6)	King Abdulaziz University (KAU) – Government Consultant
25 (I-C-7)	King Abdulaziz University (KAU)
26 (I-C-8)	Almaarefa Colleges
27 (I-C-9)	Water Research Centre

repeatedly underlined by respondents from all the expert groups interviewed, for example: *"It is unclear who is in charge of monitoring the application and adherence of the industrial facilities with Article 12, for example. Is it GAMEP? Or those licensing entities? In all cases, I can assure you that GAMEP does not have either sufficient material or human resources to follow up with this mandate properly"*. (I-A-3).

The direct consequence of such ambiguity is that this central provision occasionally remains unfulfilled, constituting a significant gap in the environmental governance system, where the operators of the regulated activities turn a blind eye to environmental obligations, as articulated by one participant:

> *"Let's be honest. We, as industrial businesses, the most important body for us is not the environmental agency but the licensing authority, which is in our case the Ministry of Commerce …Without their licence we cannot operate and thus the entire business and our commercial interests will be at stake… However, they do not too much focus on … how we operate our business environmentally. In practice, the main focus is on non-environmental affairs, such as combating commercial fraud and the quality of the final product …. And if our industry cares about the environment more than others, we will incur upon ourselves more costs giving our competitors a comparative advantage. We wish, but there is no motivation to be more environmental-friendly more than others!" (1-B-6)*.

4. Vision 2030 and the environmental challenge for KSA

4.1. Introduction

As explained earlier, Vision 2030 is a master plan to revitalise the economy, and therefore society, in multiple dimensions, including a massive move towards a low carbon economy. Closer examination shows that environmental protection considerations are highlighted and the Vision is based on the idea of aligning economic development with environmental protection, however the former seems to be a main driver.

Moreover, some ostensibly conflicting issues need to be considered. For example, the principal missions of independent institutions such as the Industrial Property Authority (MODON), and Royal Commission for Jubail and Yanbu (RCJY) are to attract national and international manufacturers and businesses. Thus, they welcome a wide range of industries, including those with potentially highly impacting industrial activities, as agreed by participants from both the bureaucracy and industry (I-A-8 and I-B-2). The degree of success is thus, inter alia, dependent on how they would combine economic and environmental aspects together. This is in line with the aims of advanced economies to "*to align efforts to secure national industrial opportunities with the development of low carbon industrial systems*" (Busch et al., 2018 p. 114).

Environmentally, Vision 2030's stated objectives appear to be similar to the longstanding sector-based and medium-centred approach to environmental governance, rather than adopting an overarching, or holistic style. Thus, there appears to be room for further development through addressing the underlying and grassroots environmental governance aspects needed to fulfil its strongly stated commitment to *harness both private sector and government funds for more efficient waste management and recycling, reducing all types of pollution, conserving water and fighting desertification"* (Vision 2030: Vibrant Society with Fulfilling Lives, 2016).

Moreover and as will be discussed below, it is expected that the various proposed developments hold the potential to bring the entire arrangement of national environmental governance to be based on the integration principle, which "*requires the effective integration of both long-term and short-term economic, environmental and social considerations in decision-making processes*" (Preston, 2016, p. 486).

4.2. Climate change and vision 2030

As stated above, the economic aspect of the Vision seems to be a principal driving force even for its initial creation. One of the main thematic goals of the Vision is to minimise the state's reliance on its petroleum assets and to maximise the sustainability of the economy. In his first interview after the introduction of Vision 2030, the Crown Prince, his Royal Highness, powerfully presented this point: "*It becomes today like our Constitution is the Quran, Sunna and the Oil. This is rather dangerous. We have become a case of oil addiction by everyone; …this has hindered the development of too many sectors in the past years*" (Al Arabiya: 33 Titles on the Vision, 2016). Thus, it should be no surprise that Vision 2030 is intended to have a direct positive effect on the issue of climate change as well as diversifying the state's national revenues. The former mainly includes establishing and investing in the clean energy sector. The Vision declares this to be a national commitment: *"Even though we have an impressive natural potential for solar and wind power, and our local energy consumption will increase three-fold by* 2030" (Vision 2030: Thriving Economy Investing for the Long Term, 2016).

Despite several effective initiatives sparked by the Vision, there are still two main areas for further improvement. The first is the need for an obligation to produce detailed and quantitatively targeted emission reduction reports by sectors and institutions to provide consistency for assessing the achieved goals. The second is for the legislative formalisation of such goals, which will both encourage the entrusted bodies to work harder to achieve the optimal level, and also subject any potential underperformance to legal investigation and/or remedy.

4.3. Different understandings of climate change

The qualitative analysis revealed surprising findings with regard to the climate change topic. The participants' overall responses, across all stakeholder categories, showed either lack of interest in this issue, or that they doubted its existence in the first place. More strikingly, they were unaware of recent international climate change developments, including the significance of the 2015 Paris

Agreement, which the KSA had signed and ratified. Within the whole array of answers, the focus was on conventional issues of environmental law, such as pollution and air quality, rather than climate change and greenhouse gases.

Even those few, including scientists, who briefly commented on the issue strongly doubted the existence and seriousness of this issue. For them, climate change and global warming issues are an overstatement of uncertain speculations by powerful international parties for purely political gains. Even the academics and scientists interviewed argued that climate changes are just natural cycles (I-C-1 and I-C-2). They also supported their unanimous view that this is a politically laden concept, by mentioning a number of proven large-scale environmental problems they perceived as much more serious, which have not gained as much momentum as the climate change issue. Hence, they shared the notion that such issues were created and powerfully advocated at the international level, driven by major powers to weaken and even blackmail emerging economies and wealthy petroleum states. Although this argument was put forward by scholars and academics and not by civil servants, its reflection on an integrated approach to tackle climate change is hardly positive.

This aggregate sceptical attitude of respondents is not easily aligned with the considerable efforts and attitudes already embraced by the formal policies and plans, including Vision 2030, in which the issue is firmly accepted, although it is often propagated indirectly, through energy efficiency and awareness campaigns. The surprise lies in the fact that the Ministry of Energy (ME) was one of the least mentioned institutions by the whole group of participants, despite its significant leadership position and its international role with regard to the issue of climate change. Even those who fleetingly mentioned the ME did not give any details or show any understanding of the nature of its role, its administrative power and capacity, and its strategy. This is in sharp contrast with their discussions about GAMEP and its roles, responsibilities, strategies and performance. This dichotomy can also be traced to the relative lack of climate change data, especially in the Arabic language, issued by the relevant institutions. The issue of the lack of material in Arabic was also mentioned by one of the respondents from the business sector (I-B-4). With the onset of Vision 2030, there should be more data available to assess environmental performance for future scholars in this area, since "embracing transparency" and ensuring availability of data is a principal component of the Vision (Effectively governed: Embracing Transparency, 2016).

5. Improving 'Integration' across law and policy to achieve vision 2030

5.1. The issue of cross-institution integration

It is well documented in the literature that environmental regulators worldwide have suffered from fragmented structures and, as a result, their integration with other institutions to achieve policy objectives is challenging (De Sadeleer, 2014). The integration principle, therefore, is considered an aspiration and at the same time a key indispensable aspect of international environmental law, as it enables it to be achieved (Birnie et al., 2009). This is further emphasised when considering energy and climate goals for the transition to a low-carbon economy; it has been noted that there is a lack of integration worldwide in terms of policy and law to achieve renewable energy development (Mauger and Barnard, 2018) and, more broadly, the energy transition (Baker et al., 2014). Further,

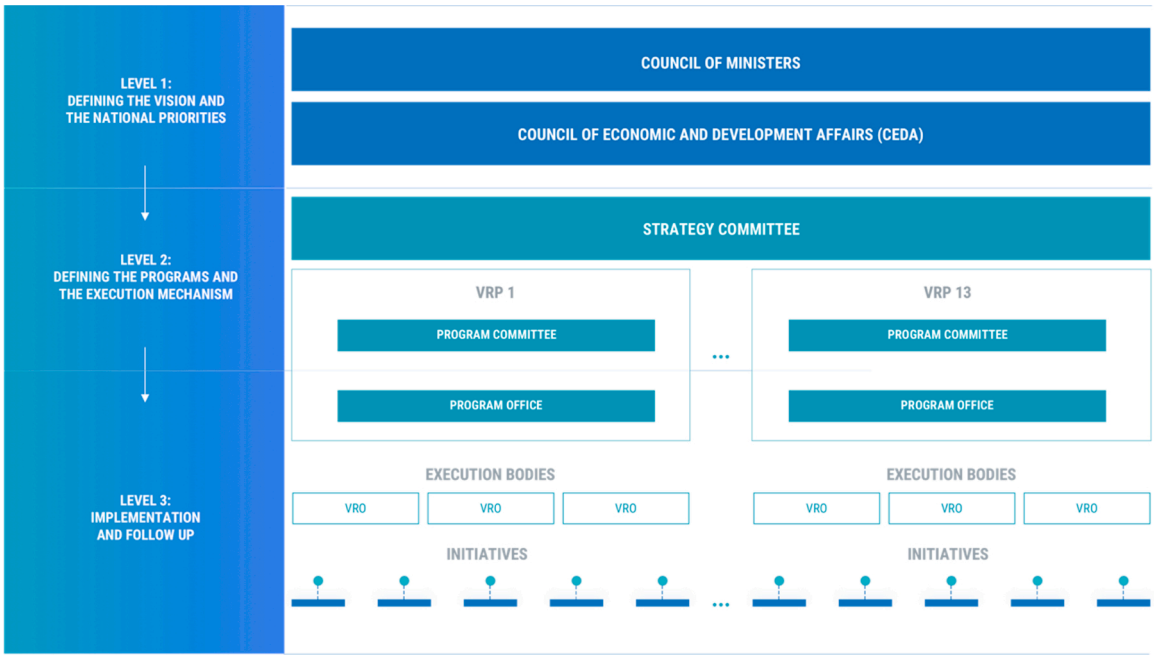

Fig. 1. The New Administrative Governance Model for Attaining Vision 2030 Aims.
Source: Vision 2030 (2016): Governance Model for Achieving Saudi Arabia's Vision 2030,

other scholars argue that this area of the transition needs scholars from environment, energy and climate change studies to unite more and work together to facilitate more integrative law and policy solutions (Heffron and McCauley, 2018).

Overall, there is lack of integration for renewable energy governance (Mauger and Barnard, 2018), and in a broader way on energy transitions (Baker et al., 2014). In addition, other transition scholars go further and argue for convergence between environmental, energy and climate change scholars (Heffron and McCauley, 2018).

This research finds similar results in the KSA's context. Indeed there is significant room for improvement within the KSA before it can achieve Vision 2030 which was launched to address challenges including those within the sphere of the environmental law and policy. Issues such as the fragmentation of environmental institutions, different understandings of and commitments towards environmental issues among the bureaucratic machines, and the lack of comprehensive environmental policies and modern enforceable environmental regulations all reduce the implementation and potential success of the integration principle, in all of its interpretations, to a more basic and fundamental level.

For example, a glaring example of such issues in the KSA is the fragmented focus by the GEL and its Rules for Implementation (RI) on the polluting emissions and discharges released into the environment. These domestic flagship environmental regulations treat land, water and air as discrete components, for example, by regulating some without the others, or regulating one in much less detail than another. Thus, the current environmental and climate governance system needs to view the environment as a single inseparable unit, in line with the core idea of the integration principle, with an appropriate understanding of the complexity and characteristics of the entire ecosystem and its ills (Sunkin and Wight, 2001; Wolf et al., 2002). These types of contextual challenges seem to have led some writers to contend that concepts such as environmental policy integration are mainly applicable to the context of domestic western politics (Biermann et al., 2009).

That being said, the KSA is aiming for reform through the new, modern "governance model" with the aim of achieving the national transformation led by Vision 2030 (Economic and Development Affairs Council, 2016). This has included, setting up the Council of Economic and Development Affairs (CEDA) and its architecture is presented below in Fig. 1. As its name suggests, CEDA is in charge of streamlining and facilitating the state's policies and goals, including environmental aims. Clear evidence of its intended environmental focus is found in the presence of the Minister of Environment, Water and Agriculture (MEWA) as an important and active member of CEDA. This new promising development is discussed in more detail in the following sections.

5.2. Developing administrative integration

As highlighted in the previous section, the administrative architecture of any state is of particular importance for the application and implementation of environmental law and policy, and the KSA is no exception. For example, until quite recently the principal environmental regulator in the KSA, i.e. GAMEP, has not been adequately equipped through law with adequate executive power to autonomously impose and enforce the environmental law (the GEL) upon the mass of regulated private and public parties. The result is that GAMEP needs to engage with a wide spectrum of executive authorities, including licensing authorities, whose cooperation or consent GAMEP needs in order to make up for its shortfall in executive power. Thus, in many cases, GAMEP has to look for a willing partner, such as another executive authority, and/or regional governments, to achieve its law and policy goals.

The repercussions of this is a lack of integration, as more often than not each of these other executive authorities has differences in understanding and potential for action towards environmental issues, and particularly if they directly interfere and overlap with economic considerations, as was discussed by participants from the bureaucracy (I-A-C and I-A-9). Moreover, these two interviewees also suggested that some regions are known to be more actively committed to environmental issues, and towards such "environmental calls for action". However, sometimes GAMEP's recommendations and requests encounter the challenges of divergent understandings and circumstances of other parts of the administrative system, resulting in a lack of action.

For the transition to a low-carbon economy to be achieved - and for the KSA this is expressed through Vision 2030 – administrative integration is vital. How this materialises will directly impact upon the way Vision 2030 to be achieved and to what extent. There are positives and negatives here. At first, there is, for example, a need to have regional application of integration policies. Until recently, GAMEP has had only regional branches in some of main cities, which means there is a lack of cross-city environmental integration. The analysis of the expert interviews also highlighted that some industries located in cities without a GAMEP branch reported that they had almost never been environmentally inspected by GAMEP. Thus, the municipality or Ministry of Municipal and Rural Affairs (MMR) attempted to cover this gap by conducting non-systematic, ad hoc and sometimes reactionary inspection visits, as explained by one of the participants from the business sector (1-B-5). Vision 2030 will optimally achieve its aims when all parts of the system work together i.e. "integration".

In light of the above, there are also positive actions happening in order to address and mitigate these challenges (lack of executive power and lack of cross-city environmental integration), for example, the inauguration in 2020 of the Special Forces for Environmental Security (SFES) (Arab News Website, 2021), a form of environmental police force to be in charge of enforcing environmental laws and regulations. Their mandate ranges from environmental surveillance, to taking preventive measures and to triggering prosecutions (Arab News Website, 2021). It is possible that this development could enhance environmental integration by unifying the enforcement of environmental legal provisions across the country, avoiding dependence on the willingness of particular executive entities to take environmentally protective actions. This is a positive development and one that will need future research to monitor its effectiveness.

5.3. Project-based integration

This point develops from the perspective of looking at general administrative development in the environmental area as the KSA

looks to transition to a low-carbon economy. The previous section focused on a more 'program' basis of integration across the system, whereas this section will focus on what happens on a project basis. This issue is a long-standing historical one in KSA (Al-Gilani and Filor, 1997 & 1999), where too often the focus is on exploring integration for environmental issues on a project-by-project basis rather than a program and system basis.

Prior to the advancement of Vision 2030, project-based levels of integration have tended to occur in the fossil fuel industries, which can be expected in an economy that is underpinned by them. In this context, environmental considerations have not always been to the fore in these instances (and in other large infrastructure projects), as indicated by interviewee I-B-I. Hence, the response has been sometimes too reactionary, and integration happens only after the fact, when it may be quite late to impact positively upon the project. Indeed, the literature highlights that merely taking environmental considerations into account and reconciling them with economic matters might even be insufficient for full integration, and that there is a need to "*give priority to environmental issues*" (Adelle and Nilsson, 2015, p.454), even over other aspects.

In this context, Vision 2030 has advanced a much needed tangible step for dealing with integration in the KSA. Since May 2018, MEWA has been taking the lead for bringing several key institutions in the same place to enhance coordination and to produce a unified national strategy named "*The National Strategy for the Environment*" – i.e. beginning with a program-based approach to action rather than the project-based approach of the past. The structure of the "Steering Committee", presided over by the Minister of MEWA, is shown in Fig. 2 below. Interestingly, the Governance Structure of this Committee includes not only environmental bodies but also energy-entrusted institutions. For instance, members of this Committee represent 20 main institutions, including the Ministry of Energy, the Saudi Energy Efficiency Center, RCJY and Ministry of Economy and Planning (Ministry of Environment, Water and Agriculture: The Executive Summary of the National Strategy for the Environment, 2020). It also embraces practitioners and scholars in multiple related disciplines, including marine environment, air quality, environmental governance and institutional framework and climate change experts. These actions taken within the KSA support the literature that identifies convergence across climate, energy and environmental issues as a key issue in achieving a transition to a low-carbon economy (Heffron and McCauley, 2018).

5.4. Developing special independent institutions

Environmental literature has long held the importance of the development of independent institutions to achieve change and this is necessary to achieve transition in the KSA. However, until quite recently (before the inauguration of Vision 2030), the KSA has encountered the challenge of too many and too few institutions at the same, with the result being there is confusion of where integration begins and where it ends. An example of this issue concerns recent developments where a new specialist independent institution, the Royal Commission for Riyadh City, has been established. As one interviewee clearly stated: "*GAMEP and other environmentally-mandated bodies are working, but discretely and in a fragmented way…. What makes the [(new) Royal Commission for Riyadh City (RCRC)][previously named] High Commission for the Development of Riyadh [HCDR] special and effective, is that it brings 17 authorities together, follows up and ensures that each one of them are progressing and achieving in accordance with the master plan of the Riyadh city …*" (I-A-9).

Thus, rather than tackle the administrative challenges posed by the existence of 17 authorities in the area to resolve environmental policy problems, a further, new institution is mandated to bring in an 'integration' policy. Nevertheless, the HCDR/RCRC has been

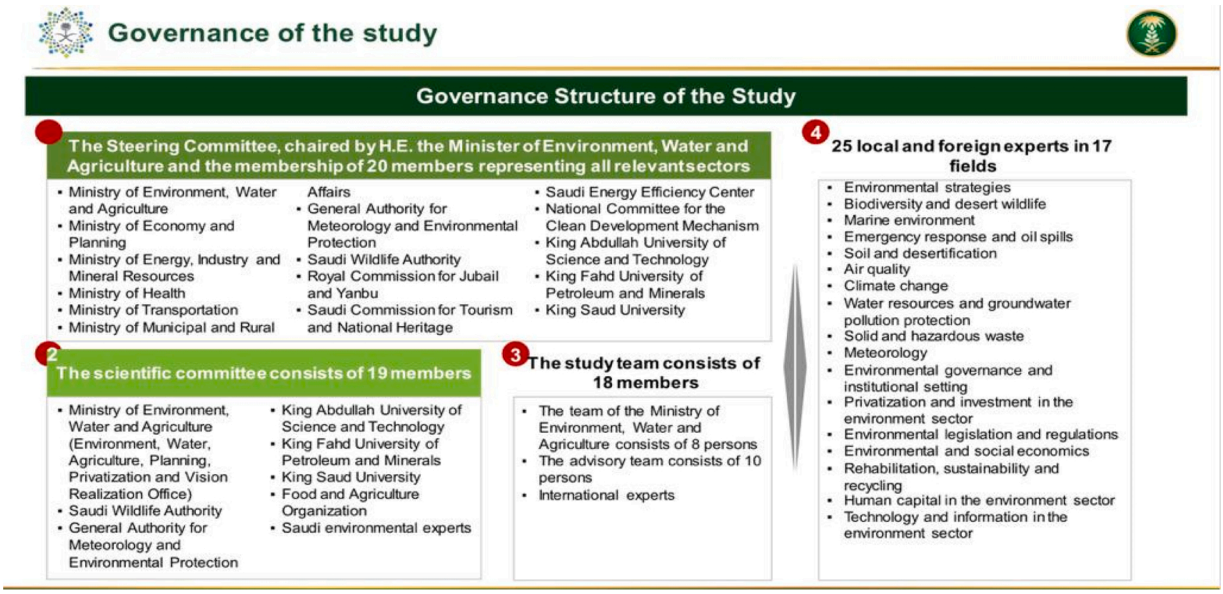

Fig. 2. The Steering Committee Structure.
Source: MEWA (National Environment Strategy: Executive Summary for the Council of Economic and Development Affairs, **2018)**

credited with achieving success for Riyadh. As a result, and in order to achieve the energy and climate change transition envisioned under Vision 2030, this policy has been replicated across the KSA. For example, the Royal Commission for AlUla (RCU) and the Higher Commission For Development of Asir Region (HCEPCR) have both been established recently. This development is consistent with the conceptual and philosophical positions which argue that overhauling the economy can be mutually supported by environmental protection considerations and which seek to disprove the perception that there is a trade-off between these aspects (Dryzek, 2013; Holder and Lee, 2007).

6. Conclusion

This paper contributes to transition study literature in emphasising the role of law in achieving a low-carbon (or sustainable) transition. The paper has built on the existing literature to examine the promising potential of the national master plan Vision 2030 on environmental governance in the KSA, including the plan to transition to a lower-carbon economy. Following the call of authors such as Köhler et al. (2019) for more practice-based research, this paper, based on a study which included interviews with key stakeholders in the KSA, has illuminated the practices that support or impede the moves to this promising transition from both a law and policy perspective.

There are a number of further key and novel contributions from this research regarding the potential of Vision 2030 to successfully deliver the transition. The analysis has identified significant progress as well as room for improvement concerning action on climate change towards the year 2030. It has been concluded that the principal environmental regulations and policy documents have not so far elaborated sufficiently on the issue of delivering on climate change policy objectives to meet Vision 2030. However, the newly introduced environmental legislations and institutions have the potential to address several long-existing challenges.

Overall, it can be concluded that climate change issues are becoming more prominent in the KSA. However, until recently, the basic regulatory and legal infrastructure and the "conventional environmental problems" have remained the key topics of importance to the public institutions. Hence, there is great potential for Vision 2030 to bridge this gap between local concerns and the current international agenda on climate change. Moreover, the qualitative analysis revealed a significant amount of uncertainty and scepticism among individual respondents regarding climate change, although not among those representing environmental institutions and several participants revealed their outlook on the issue as politically driven by international organisations, rather than based on an environmentally-centred perspective. This was in contrast to the prominent efforts exerted, notably by the ME, to interact with the international community to address climate change. Fortunately, with the advent of Vision 2030, the rhetoric of climate change is now more in line with the desired economic reforms.

Another central finding is the need to further promote the principle of integration in both tackling climate change and to address other more conventional environmental challenges, and hence to achieve a just transition to a low-carbon economy. The issues identified in this paper demonstrate how the integration of environmental factors and a common understanding of climate change issues are important to further facilitate the realisations of the aims of Vision 2030. Nevertheless, Vision 2030 is already achieving change and encouraging more integration across law, policies, institutions and across the regions. There are clear plans for new strategies, enforcement of regulations and feedback mechanisms to deliver on environmental performance. However, further developments will be needed, particularly in terms of improved monitoring and reporting mechanisms that allow for feedback, follow-up, restorative action and environmental accountability to ensure the plan reaches its optimal potential and success by 2030.

The research has also identified variations in understanding and prioritising climate change and environmental issues across and within institutions, as well as a need to increase their executive capacity for a transition to a lower-carbon economy. For example, until the relatively recent introduction of Vision 2030, the principal environmental regulations have concentrated on particular environmental activities and have not thoroughly regulated all offending industrial activities; too often there is project focus rather than program focus for reform. In this respect environmental legislative gaps have been identified in this study, together with the legal implications of leaving some environmental tasks unfulfilled and with no alignment of responsibility. This combination of factors results in sub-optimal environmental accountability mechanisms and, inevitably, substandard environmental outcomes. This issue of increased accountability is a key aim of Vision 2030, and hence, reform is on the agenda and expected to be undertaken on this issue.

A further key issue is the need for considerable expansion of expert capacity to ensure that Vision 2030 can be delivered in terms of the environmental challenges. This is reflected in variation of understanding, performance and decisions by environmental inspectors, as discussed above. For example, there is a need for improved relationships between environmental scientists and bureaucracy, as sub-optimal relationships have contributed to insufficient integration of environmental science into environmental-decision making. In a similar vein, different prioritisations of environmental considerations among institutions contribute to the lack of emphasis on and implementation of the integration principle among both publicly- and privately-owned industrial development activities. This results in relatively less consideration of environmental issues in comparison to economic and financial considerations, which have tended to be awarded higher levels of established importance across many projects in the KSA prior to the advent of Vision 2030.

Moreover, there has been a persistent fragmentation of environmentally-mandated bodies. It has been shown how the existence of parallel national and local environmental regulators has brought about different issues regarding integration and coordination, which have, in turn, contributed to variation in environmental protection efforts across different areas. Another challenge which has been identified relates to the distribution of complex executive powers for environmental matters. However, Vision 2030 has started addressing such challenges, including through the amalgamation of some environmental institutions into one large Ministry of Environment (MEWA).

Finally, this research has highlighted the lack of presence of the environmental principle of integration, despite its significance in any environmental governance system and ability to achieve the kind of major changes aimed at to fulfil the ambitious national master

plan laid out in Vision 2030. Nevertheless, commitment of Vision 2030 to the institutionalisation of this principle is an encouraging and needed development. It is also clear that the developments and concerns identified in this study do not appear to be sufficiently addressed by the existing literature on the KSA or within practice, as evidenced by the interviews conducted with expert stakeholders. Further research is needed to address more of these issues within the Middle Eastern context, and to discover where issues of energy, environment and climate change converge to affect whether and how different countries will achieve their just transition to a low-carbon economy.

Declaration of Competing Interest

The authors report no declarations of interest.

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Appendix A

Selection of Interviewees

Participants were chosen on the basis of their perceived knowledge and/or practical involvement in environmental protection circles and the environmental governance field in general. This required first identifying the main authorities and institutions which have a clear involvement in the dynamics of environmental protection. Based on the types of institutions identified, interviewees were divided into four categories as:

- Bureaucrats or civil servants (I-A)
- Representatives from industries and the private sector (I-B)
- Academics (I-C)

Nevertheless, representatives from each of the principal environmental bodies were interviewed. These institutions included the Ministry of Environment, Water and Agriculture (MEWA) and the environment agency (GAMEP), the Ministry of Municipal and Rural Affairs (MMRA), the Royal Commission for Jubail and Yanbu (RCJY) – Environmental Protection Department and the Saudi Industrial Property Authority (MODON) Environmental Protection Department, as well as representatives from major national industries and companies, in addition to professors from principal universities in the KSA. Thus, the interview process encompassed participants from a broad range of specialisations or fields of scholarship, including Environmental Science, Environmental Engineering, Petroleum Engineering, Biological Science, Nanotechnology and Toxicology, Civil Engineering, Environmental International Relations.

However, there was limited opportunity to interview professional environmental lawyers as opposed to academic lawyers, as this specialisation in practice is limited and there were none obtained in this area despite searches. It is hoped that this research may contribute to highlighting this under-researched area of knowledge and attract the attention of researchers to conduct future studies in the context of KSA or its regional neighbours.

The Questions Asked

The interviewees were asked a total of 35 questions, classified under the following themes: climate change, environmental governance modes, environmental governance and risks; environmental principles; good environmental governance; scales or levels of environmental governance; trends in environmental governance; environmental governance and societal actors; environmental governance and the unique characteristics of environmental problems.

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